

LESSON GUIDE

Technology Plus

Unit 3: Hardware

2.5.1 – Input Output Interfaces

Lesson Overview:

In this lesson, students will explore how wired and wireless interfaces make it possible for devices to communicate and share data.

Students will:

- Distinguish between wired and wireless interfaces and explore the different options of each interface.
- Identify different networking tools.
- Apply their knowledge of wired and wireless interfaces to determine which best fits a user's needs.

Guiding Question: How do wired and wireless interfaces impact networking?

Suggested Grade Levels: 8-12

Technology Needed: No

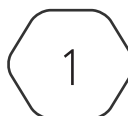
Approximate Time Required: 45-70 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.5 – Compare and contrast common types of input/output device interfaces.
 - Networking
 - Wired
 - Ethernet connector (RJ45)
 - Fiber connector small form-factor pluggable (SFP)
 - Wireless
 - Bluetooth
 - Near-field communication (NFC)
 - 802.11X
 - Networking devices and tools
 - Crimpers
 - Cable testers

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Lesson Background

Background Information:

In today’s world, devices rely on a variety of interfaces to communicate, each serving a distinct purpose. From Ethernet – reliable, wired, high-speed data transfer – to wireless connections like Bluetooth and Wi-Fi, these technologies shape how we work, share, and interact every day. Students will examine the most common networking interfaces, and the tools professionals use to configure them. Networking underpins nearly all device functionality; without wired or wireless interfaces, modern computing would be severely limited.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.5.1 – Input Output Interfaces • Activity Slides 2.5.1 – Wired vs Wireless Interfaces
Materials per Student	• Student Notes 2.5.1 – Input Output Interfaces • Student Handout 2.5.1 – Input Output Interfaces • Assessment 2.5.1 – Input Output Interfaces (optional)

Lesson Background

Cyber Terms:

Term	Definition
Ethernet connector	a standard interface for connecting computers and network devices in a wired local area network, also called registered jack 45 (RJ45)
small form-factor pluggable (SFP)	connectors used for network interfaces in fiber optic communications
Bluetooth	a short-range wireless technology used for connecting devices (typically up to 33 feet) without physical cables, often used in battery-powered devices
near-field communication (NFC)	a short-range wireless technology used for communication between devices that are very close together
802.11X	a set of standards for wireless networking, referred to as Wi-Fi, used to connect devices over a wireless local area network
crimpers	tools used to attach connectors to the end of network cables
cable testers	devices used to test the functionality and integrity of network cables

2.5.1 – Input Output Interfaces

Guiding Question: How do wired and wireless interfaces impact networking?

Lesson Launch (5-10 minutes)

- Have students imagine that the Wi-Fi in their home has suddenly stopped working.
- **Ask:** “How would you stay connected? Would you switch to a wired Ethernet connection? Use your phone’s hotspot? Rely on Bluetooth?”
- Give students time to think and answer.
- There is no right or wrong answer, students should focus on how they would stay connected and why they would choose that way.
- **Say:** “Networking interfaces allow a computer or device to connect to a network so it can send and receive data with other devices. Some network devices use physical cables, while others use radio signals. Understanding the difference between wired and wireless interfaces is essential for building, maintaining, and troubleshooting networks.”

Networking Interfaces and Tools (20-25 minutes)

- Distribute Student Handout 2.5.1 – Input Output Interfaces and encourage students to fill in the answers during the lesson.

Wired Networking Options

- Introduce an **Ethernet connector**, its definition, and discuss its features:
 - They transmit data via copper pairs.
 - Within each pair, one wire is primarily white with a colored stripe, while the other wire is a solid color.
 - The color codes ensure the wires are aligned correctly in the connector, allowing the cable to function properly.
 - These two standards EIA/TIA-568A/B create a universal language for Ethernet wiring.
- Introduce a **small form-factor pluggable connector (SFP)**, its definition, and discuss its features:
 - SFPs are a type of fiber optic cable.
 - Fiber optic cables work by using light to carry information and provide significantly higher throughput rates over longer distances than copper cables.
 - Fiber connector SFPs can support very high data transfer rates and allow for long-distance communication with minimal signal loss.

Wireless Networking Options

- Introduce **Bluetooth**, its definition, and discuss its features:
 - Short-range (up to 33 feet)
 - Low power, designed for battery-operated devices
 - Slower transfer speeds than Wi-Fi

- Introduce **near-field communication (NFC)**, its definition, and discuss its features.
 - Very short range (a few cm)
 - Often used for payments or secure data exchange
 - Uses minimal power
- Introduce **802.11X**, its definition, and discuss its features:
 - Supports high-speed data transfers, with speed depending on the standard.
 - Covers larger areas than Bluetooth and NFC, but range can be reduced by walls or interference.
 - Requires strong security measures to protect against unauthorized access.
 - Wi-Fi is not an abbreviation for anything. Students may think it stands for Wireless Fidelity.

Networking Tools

- Introduce a **crimper**, its definition, and discuss its features.
 - Used to attach connectors (like RJ-45) to network cables, allowing for custom cable lengths and configurations.
 - Require skill and practice to ensure proper connections; often paired with a cable tester to verify accuracy.
- Introduce a **cable tester**, its definition, and discuss its features.
 - A cable tester is used to verify that a network cable is wired correctly and functioning as intended.
 - It can quickly detect wiring faults, short circuits, or mis-pinned connections, ensuring cables meet performance standards.
 - Note that cable testers only evaluate the cable itself – they don't measure network speed or diagnose issues with connected devices.
- Students learn more about these concepts in Unit 4: Networking.

Activity: Wired vs. Wireless Interfaces (15-25 minutes)

- Display Activity Slides 2.5.1 – Wired vs. Wireless Interfaces and direct students to record their answers on the bottom of the handout.
- Decide if students will work individually or in small groups.
- Read each scenario aloud (or allow students/groups to read independently).
- Students determine whether a wired or wireless interface fits the needs, specifying the exact type of connection when possible.
- Encourage consideration of speed, mobility, reliability, and security when making decisions.
- Allow time for students to write their answers.
- Remind students that answers may vary, as long as they are justified and reasonable.

Scenarios and Answers

- Alex just built a high-performance gaming PC in his bedroom and wants the fastest and most stable connection for competitive online gaming.

Alex should have a wired Ethernet connection. Ethernet connections provide consistent speeds, low latency, and eliminate the possibility of interference from other devices.

- Sarah is setting up a new corporate office for 50 employees, each with a desktop computer and Voice over Internet Protocol (VoIP) phones. The company also provides laptops for mobile work and wants to support guests on the network.

Sarah's office could benefit from a hybrid setup. Wireless interfaces would allow laptops and guests to connect to the network without cables and allow for mobility. Wired interfaces would provide high speed and reliability for fixed desktops and VoIP phones.

- Chris is upgrading his home with smart lights, smart locks, and a video doorbell. He wants everything connected to a central hub for easy control from his phone.

Chris needs a wireless interface because IoT devices need mobility and convenience. IoT devices can connect using Wi-Fi and Bluetooth, eliminating the need for cables running through the home.

- Dr. Patel is deploying a patient monitoring system in a hospital where critical data like heart rate and oxygen levels must be sent to nurses' stations in real time.

Dr. Patel's system would function best with a wired small form-factor pluggable fiber connection. SFPs are reliable, fast, and secure. They protect against electromagnetic interference, providing data integrity. This wired connection will also prevent signal interference from other hospital equipment. SFPs can be costly but may be worth the price for a medical business.

- Emma and her friend want to quickly share photos and a short video between their smartphones while at a coffee shop.

Emma and her friend should use a wireless near-field communication interface to share files. This will allow them to share their files with the click of a button. NFC will work best for small files and because the friends are close together.

Putting It All Together (5-10 minutes)

- Review what students learned today by discussing the questions below as a class.
- Why might someone prefer a wired connection over a wireless connection?

Wired connections are faster, have more stable performance, have less interference from other devices, and are more secure due to the requirement of physical access.

- Why might someone prefer a wireless connection over a wired connection?

Wireless connections allow for mobility and convenience when connecting devices, work with portable devices that do not have Ethernet ports and have easier installation due to no cables.

- How are crimpers and cable testers beneficial when setting up network devices?

Crimpers allow technicians to attach connectors to the end of Ethernet cables to allow for custom cable lengths and configurations. Cable testers verify that cables are correctly wired and fully functional, preventing connection failures and troubleshooting issues.